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Effects of amycenone on serum levels of tumor necrosis factor-α, interleukin-10, and depression-like behavior in mice after lipopolysaccharide administration

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ABSTRACT

Accumulating evidence suggests that inflammation plays a role in the pathophysiology of depression and that anti-inflammatory substances have antidepressant effects. Amycenone is obtained from extracts of the Yamabushitake (*Hericium erinaceum*). The purpose of this study is to examine whether amycenone shows anti-inflammatory and antidepressant effects in an inflammation-induced mouse model of depression. First, we examined the effects of amycenone on the serum levels of the pro-inflammatory cytokine, tumor necrosis factor-α (TNF-α), and the anti-inflammatory cytokine, interleukin-10 (IL-10), after intraperitoneal administration of the bacterial endotoxin lipopolysaccharide (LPS). Oral administration of amycenone (50, 100, or 200 mg/kg) markedly blocked an increase in the serum TNF-α levels after a single administration of LPS (0.5 mg/kg). Furthermore, amycenone (200 mg/kg) markedly increased the serum IL-10 levels by a single administration of LPS (0.5 mg/kg). Next, we examined the effects of amycenone on depression-like behaviors in the tail-suspension test (TST) and forced swimming test (FST). Pretreatment with amycenone (200 mg/kg) significantly attenuated LPS (0.5 mg/kg)-induced increase of the immobility time by the TST and FST, indicating antidepressant effects of amycenone. In addition, oral administration of paroxetine (30 mg/kg) showed anti-inflammatory and antidepressant effects in the same model. These findings suggest that amycenone has antidepressant effects in LPS-induced inflammation model of depression. Therefore, amycenone could represent a potential supplement to prevent inflammation-related depression.

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1. Introduction

Depression is one of the most common psychiatric disorders and the leading cause of disability worldwide. The World Health Organization estimates that more than 350 million individuals of all ages suffer from depression (WHO, 2012). Therefore, depression is a major contributor to the global burden of disease (Walker et al., 2015; Whiteford et al., 2013). In the view of the impact of mental health, new approaches need to be developed to prevent or delay the progression of depression. Accumulating evidence suggests that nutrition influences the development of numerous psychiatric disorders, including depression (Bazinet and Layé, 2014; Chen et al., 2012; El-Behadli et al., 2015; Hashimoto, 2014; Mello et al., 2014; Murakami and Sasaki, 2010; Opie et al., 2014; Shirai et al., 2012). A recent meta-analysis including 21 studies demonstrated that high intake of fruit, vegetables, fish, and whole grains are associated with a reduced risk of depression (Lai et al., 2014). Taken together, nutrition plays an important role in mental health.

Inflammation also plays a role in the pathophysiology of depression, and some anti-inflammatory drugs have antidepressant-like effects (Hashimoto, 2009, 2015; Miller et al., 2009; Raison et al., 2010; Dantzer et al., 2008). Alternatively, when bacterial endotoxin lipopolysaccharide (LPS) is administered to rodents, depression-like behaviors are induced after the induction of inflammation (Dantzer et al., 2008; O'Connor et al., 2009). Furthermore, antidepressants, including selective serotonin reuptake inhibitors (SSRIs) and serotonin and norepinephrine reuptake inhibitors (SNRIs), could prevent depression-like behavior and alternations in serum pro-inflammatory cytokines, such as tumor necrosis factor-α (TNF-α), caused by LPS administration (Ohgi et al., 2013). A meta-analysis showed the presence higher serum TNF-α levels in drug-free depressive patients compared with those in healthy controls (Dowlati et al., 2010). A study using postmortem brain samples

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showed elevated gene expression of pro-inflammatory cytokines in the frontal cortex of people with a history of depression (Shelton et al., 2011). Thus, it is likely that both peripheral and central inflammation is associated with depression and that anti-inflammatory drugs could ameliorate depressive symptoms in depressed patients.

Lion's Mane Mushroom (*Hericium erinaceum*) is a popular edible mushroom in East Asian countries, including Japan, Korea, and China, and it is utilized in traditional Chinese medicine. The mushroom is called "Yamabushitake" and "Houtou" in Japan and China, respectively. The fruiting bodies of *H. erinaceum* contain several bioactive substances, including aromatic compounds, diterpenoids, steroids, and polysaccharides (Jiang et al., 2014; Khan et al., 2013; Wasser, 2014). A double-blind, placebo-controlled study showed that oral administration of dried Yamabushitake powder to Japanese subjects is effective for the treatment of mild cognitive impairment patients (n = 15) (Mori et al., 2009).

Amycenone is a *H. erinaceum* extract standardized to contain hericenones (0.5%) and amyloban (6%) (Inanaga, 2012). Amyloban®3399 is a finished product made from amycenone. Inanaga et al. reported that Amyloban® 3399 is effective in the treatment of cognitive impairment in recurrent depressed patient (n = 1) (Inanaga, 2014) and treatment-resistant patients (n = 10) with schizophrenia (Inanaga et al., 2014; Inanaga, 2015). However, the precise mechanisms underlying its therapeutic effect are currently unknown. The purpose of this study is to examine whether amycenone shows anti-inflammatory and antidepressant effects on depression-like behaviors in mice after LPS administration. First, we examined the anti-inflammatory effects of amycenone on LPS-induced inflammation, by measuring the changes in the serum levels of pro-inflammatory cytokine, TNF-α, and the anti-inflammatory cytokine, IL-10, in mice. Second, we evaluated the antidepressant effect of amycenone on LPS-induced depression-like behavior in mice.

2. Materials and methods

2.1. Animals

Male adult C57BL/6N mice (body weight 20–28 g, Japan SLC, Inc., Hamamatsu, Japan) were used for all experiments at age 8 weeks. Animals were housed under controlled temperature and 12 h light/dark cycles (lights on between 07:00–19:00), with ad libitum food and water. All experiments were carried out in accordance with the Guide for Animal Experimentation of Chiba University.

2.2. Drug administration

On the day of injection, fresh solutions were prepared by dissolving compounds in sterile endotoxin-free isotonic saline. Lipopolysaccharide (0.5 mg/kg; L-4130, serotype 0111:B4, Sigma-Aldrich, St Louis, MO, USA) was administered intraperitoneally (i.p.). Amycenone was prepared at Mushroom Wisdom Ltd. (NJ, USA) and was dissolved in 0.5% carbomethoxycellulose (CMC; Wako Pure Chemical Industries, Ltd., Tokyo, Japan). Amycenone is a commercial material made from Lion's Mane extracts and powderized by mixing with Lion's Mane powder, which is standardized to contain 0.5% hericenones and 6% amyloban. The dose of amycenone was selected, as reported previously (Zhuang et al., 2009). Paroxetine hydrochloride (Wako Pure Chemical Co., Tokyo, Japan) was dissolved in distilled water, and the dose (30 mg/kg) was selected, as reported previously (Ohgi et al., 2013).

2.3. Enzyme-linked immunosorbent assay (ELISA)

Vehicle (10 ml/kg; 0.5% CMC) or amycenone (50, 100 or 200 mg/kg) were administered orally to mice 60 min before LPS (0.5 mg/kg, i.p.) injection. Under sodium pentobarbital, blood samples were taken via a cardiac puncture 90 min after LPS administration. Blood was centrifuged at 2000 *g* for 20 min to generate serum samples, as previously reported (Ma et al., 2014; Ohgi et al., 2013). Serum samples were diluted 10-fold with ELISA diluent solution (eBioscience Inc., San Diego, CA, USA). Serum concentrations of TNF-α and IL-10 were measured using a Ready-SET-Go ELISA kit (eBioscience Inc., San Diego, CA, USA), according to manufacturer's instructions.

2.4. Behavioral tests

In order to examine the effects of amycenone on LPS-induced depression-like behaviors, amycenone (200 mg/kg) was pretreated 60 min prior to LPS injection. Behavioral evaluations were performed 24 h after LPS injection. On day 1 (9:00 AM), vehicle (10 ml/kg; 0.5% CMC) or amycenone (200 mg/kg) (or paroxetine (30 mg/kg)) was orally administered. One hour later, saline (10 ml/kg) or LPS (0.5 mg/kg) was administered intraperitoneally (i.p.) into mice. On day 2, all behavioral tests were performed in the following order: locomotion on 9:00 AM (24 h after LPS injection), tail suspension test (TST) on 14:00 PM (27 h after LPS injection), forced swimming test (FST) on 16:00 PM (29 h after LPS injection), as reported previously (Ma et al., 2014; Zhang et al., 2015). Mice were put into the test room 30 min before the behavioral tests. All tests were performed between 9:00 AM–17:00 PM, in a quiet room. After over all test, mice were replaced in to their home cages and returned to breeding room. The FST and TST in mice are the most widely used behavioral assays for detecting potential antidepressant-like activity (Cryan and Holmes, 2005; Ma et al., 2011, 2013, 2014; Zhang et al., 2015).

2.4.1. Locomotion

Locomotion was measured using the SCANET MV-40 (MELQUEST Co., Ltd., Toyama, Japan). This system consists of a black square cage (565 × 565 mm) equipped with two crossing sensor frames of 72 (x axis) × 72 (y axis) pairs of near-infrared beam sensors, spaced 6 mm apart and at right angles to each other. The beam sensors thus form two parallel horizontal grids set at different heights. A transparent Plexiglas cage (560 × 560 × 330 mm) for the mouse was centered inside the system. Each pair of sensors was scanned every 0.1 s to detect animal movement. Two different variables of horizontal movements could be monitored by the lower sensors: small horizontal movements of 12 mm or more (M1; 1 U = 6 mm) and large horizontal movements of 60 mm or more (M2; 1 U = 6 mm). The locomotion was been performed under house lighting and the movement of the cumulative distance was recorded for 60 min. The cage was cleaned between testing session.

2.4.2. Tail suspension test (TST)

The mice were taken from their home cage and a small piece of adhesive tape was placed approximately 2 cm from the tip of the tail. A single hole was punched in the tape and the mice were hung individually on a hook. The immobility time of each mouse was recorded 10 min. All mice were videoed and then the immobility time of tail suspension was recorded by a skilled observer who used stopwatch. Mice were considered immobile only when they hung passively and completely motionless.

2.4.3. Forced swimming test (FST)

The mice were placed individually in a cylinder (diameter: 23 cm; height: 31 cm) containing 15 cm of water maintained at 23 ± 1 °C. The animals were not able to touch the bottom. Animals were tested in an automated forced-swim apparatus by the SCANETMV-40 (MELQUEST Co., Ltd., Toyama, Japan). The cylinder was placed in a box with infrared cell sensors on the walls to detect swimming activity. The software set up a rectangle that circumscribed the body of an animal every 0.3 s. If the animals were out of the rectangle (i.e., a part of the animal body was detected outside the rectangle) 0.3 s after

setting up the rectangle, the software counted as "movement (swimming)" in that period. "Immobility" was defined as such if the animal stayed within the same rectangle 0.3 s after setting up the rectangle. Immobility time was calculated from activity time as (total)–(active) time by using analysis software of the apparatus. Cumulative immobility time was scored for 6 min during the test.

2.5. Statistical analysis

The data are shown as the mean ± standard error of the mean (S.E.M.). Data analysis was performed using the PASW Statistics 20 (formerly SPSS statistics; SPSS, Tokyo, Japan). The data were analyzed using a one-way analysis of variance (ANOVA) and two-way ANOVA, followed by post-hoc Fisher's least significant difference (LSD) test or the unpaired t-test. p values of less than 0.05 were considered statistically significant.

3. Results

3.1. Anti-inflammatory effects of amycenone on serum levels of TNF-α and IL-10 after LPS administration

Amycenone (50, 100, or 200 mg/kg) was pretreated 60 min prior to LPS injection, and blood was collected 90 min after LPS injection (Fig. 1A). In vehicle-treated mice, serum levels of TNF-α were very low (Fig. 1B), consistent with the previous report (Ma et al., 2014; Ohgi et al., 2013). Serum levels of TNF-α were significantly increased after a single administration of LPS (0.5 mg/kg) (Fig. 1B). One-way ANOVA analysis of TNF-α data revealed significant differences among the six groups [$F_{(5, 66)} = 10.039$, $p < 0.001$]. Pretreatment with amycenone (50, 100, or 200 mg/kg) significantly attenuated LPS-induced increases in serum TNF-α (Fig. 1B).

In vehicle-treated mice, serum levels of IL-10 were low. Serum levels of IL-10 were significantly increased after a single administration of LPS (0.5 mg/kg) (Fig. 1C). One-way ANOVA analysis of IL-10 data revealed significant differences among the five groups [$F_{(4, 34)} = 8.572$, $p < 0.001$]. Pretreatment with amycenone (200 mg/kg) significantly increased serum levels of IL-10 after LPS administration (Fig. 1C).

3.2. Antidepressant effects of amycenone on LPS-induced depression-like behaviors in mice

Amycenone (200 mg/kg) was pretreated 60 min prior to LPS injection, and behavioral tests were performed 24 h after LPS injection (Fig. 2A). Two-way ANOVA of locomotion data revealed no significant differences [LPS: $F_{(1, 44)} = 0.009$, $p = 0.927$, amycenone: $F_{(1, 44)} = 0.12$, $p = 0.730$, interaction: $F_{(1, 44)} = 0.135$, $p = 0.715$]. Pretreatment with amycenone did not affect the spontaneous locomotion in the vehicle-treated and LPS-treated mice (Fig. 2B). Two-way ANOVA of TST data revealed significant differences [LPS: $F_{(1, 44)} = 14.089$, $p = 0.001$, amycenone: $F_{(1, 44)} = 4.827$, $p = 0.034$, interaction: $F_{(1, 44)} = 1.906$, $p = 0.175$]. Post hoc analysis showed that amycenone (200 mg/kg) significantly attenuated the immobility time compared with the LPS-treated group (Fig. 2C). Two-way ANOVA of FST data revealed significant differences [LPS: $F_{(1, 44)} = 4.171$, $p = 0.048$, amycenone: $F_{(1, 44)} = 7.424$, $p = 0.009$, interaction: $F_{(1, 44)} = 1.670$, $p = 0.204$]. Post hoc analysis showed that amycenone (200 mg/kg) significantly attenuated the immobility time compared with the LPS-treated group (Fig. 2D). However, amycenone (200 mg/kg) alone did not affect the immobility time (TST and FST) of control mice (Fig. 2 C and D).

3.3. Anti-inflamamtory effects of paroxetine on serum levels of TNF-α and IL-10 after LPS administration

Paroxetine (30 mg/kg) was pretreated 60 min prior to LPS injection, and blood was collected 90 min after LPS injection (Fig. 3A). Two-way ANOVA of TNF-α data revealed significant differences [LPS: $F_{(1, 31)} = 42.998$, $p < 0.001$, paroxetine: $F_{(1, 31)} = 10.959$, $p = 0.003$, interaction: $F_{(1, 31)} = 6.025$, $p = 0.021$]. Pretreatment with paroxetine (30 mg/kg) significantly attenuated LPS-induced increases in serum TNF-α (Fig. 3B). Two-way ANOVA of IL-10 data revealed significant differences [LPS: $F_{(1, 31)} = 153.183$, $p < 0.001$, paroxetine: $F_{(1, 31)} = 73.5$, $p < 0.001$, interaction: $F_{(1, 31)} = 71.803$, $p < 0.001$]. Pretreatment with paroxetine

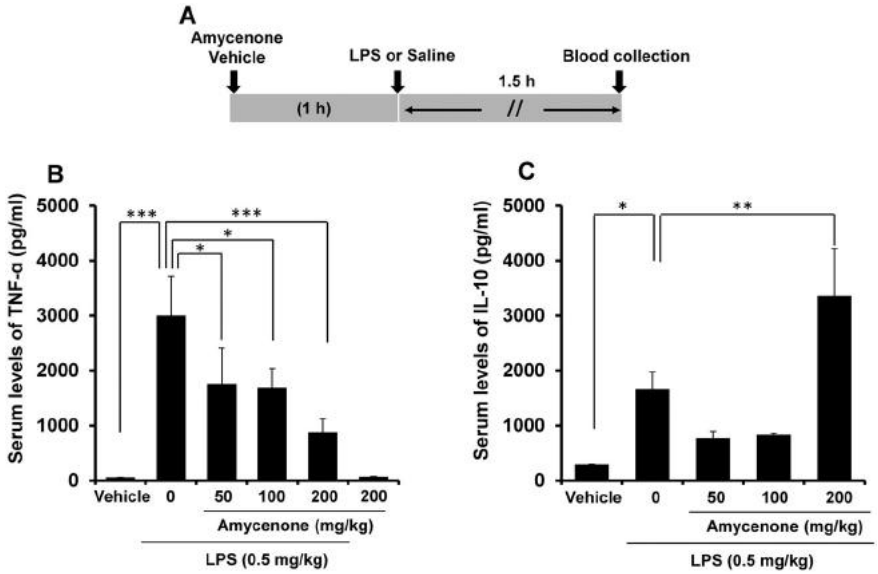

Fig. 1. Anti-inflammatory effects of amycenone on LPS-induced increase in serum levels of TNF-α and IL-10. Sixty minutes after a single oral administration of vehicle (10 ml/kg), or amycenone (50, 100, or 200 mg/kg), LPS (0.5 mg/kg, i.p.) or saline (10 ml/kg, i.p.) was injected. Blood sample was collected 90 min after LPS injection. Serum levels of TNF-α and IL-10 were measured with ELISA. A: Schedule of treatment and sample collection, B: TNF-α, C: IL-10. Data are shown the mean ± S.E.M. (n = 12 for TNF-α, n = 7 or 8 for IL-10). *p < 0.05, **p < 0.01, ***p < 0.001 compared with LPS-treated group.

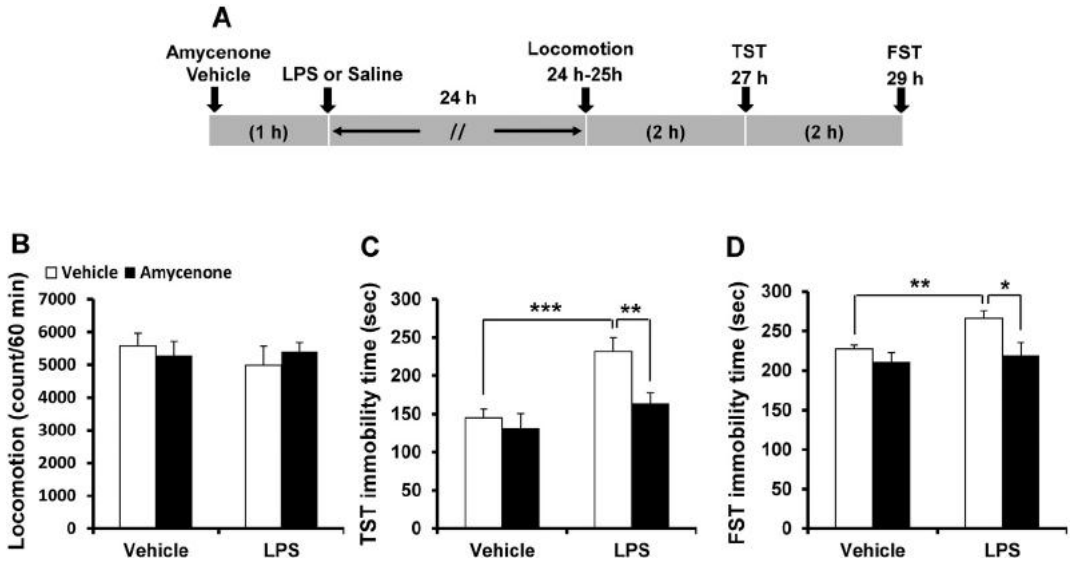

Fig. 2. Antidepressant effects of amycenone on LPS-induced depression-like behaviors in mice. Amycenone (200 mg/kg) or vehicle (10 ml/kg) was administered orally 60 min before LPS (0.5 mg/kg, i.p.) administration. Behavioral evaluations were performed 24 h after LPS injection. A: Schedule of treatment and behavioral tests. On day 1, vehicle (10 ml/kg) or amycenone (200 mg/kg) was orally administered. One hour later, saline (10 ml/kg) or LPS (0.5 mg/kg) was administered i.p. into mice. On day 2, all behavioral tests were performed in the following order: locomotion on 9:00 AM (24 h after LPS injection), tail suspension test (TST) on 14:00 PM (27 h after LPS injection), and forced swimming test (FST) on 16:00 PM (29 h after LPS injection). B: locomotion, C: tail-suspension test (TST), D: forced-swimming test (FST). The values represent the mean ± S.E.M. (n = 11–12). *p < 0.05, **p < 0.01 and ***p < 0.001 compared with the vehicle + LPS group.

(30 mg/kg) significantly increased serum levels of IL-10 after LPS administration (Fig. 3C).

3.4. Antidepressant effects of paroxetine on LPS-induced depression-like behaviors in mice

In order to examine the effects of the SSRI paroxetine on LPS-induced depression-like behaviors, paroxetine (30 mg/kg) was pretreated 60 min prior to LPS injection (Fig. 4A). Behavioral evaluations were performed 24 h after LPS injection. Two-way ANOVA of locomotion data revealed no significant differences [LPS: $F_{(1, 45)} = 0.169$, $p = 0.683$, paroxetine: $F_{(1, 45)} = 0.498$, $p = 0.484$, interaction: $F_{(1, 45)} = 0.472$, $p = 0.496$]. Pretreatment with paroxetine did not affect the spontaneous locomotion in the vehicle-treated and LPS-treated mice (Fig. 4B). Two-way ANOVA of TST data revealed significant differences [LPS: $F_{(1, 45)} = 10.257$, $p = 0.003$, paroxetine: $F_{(1, 45)} = 8.747$, $p = 0.005$, interaction: $F_{(1, 45)} = 3.506$, $p = 0.068$]. Post hoc analysis showed that paroxetine significantly attenuated the immobility time compared with the LPS-treated group (Fig. 4C). Two-way ANOVA of FST data revealed significant differences [LPS: $F_{(1, 44)} = 4.326$, $p = 0.044$, paroxetine: $F_{(1, 44)} = 4.735$, $p = 0.035$, interaction: $F_{(1, 44)} = 1.371$, $p = 0.248$]. Post hoc analysis showed that paroxetine significantly attenuated the immobility time compared with the LPS-treated group (Fig. 4D). However, paroxetine (30 mg/kg) alone did not affect the immobility time (TST and FST) of control mice (Fig. 4C and D).

4. Discussion

In the present study, we demonstrated that amycenone has anti-inflammatory effects on the serum levels of TNF-α after LPS injection

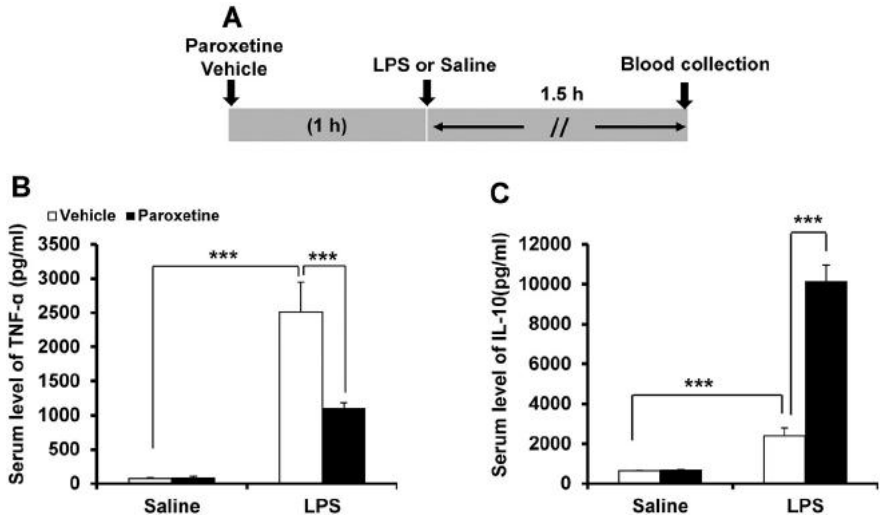

Fig. 3. Anti-inflammatory effects of paroxetine on LPS-induced increase in serum levels of TNF-α and IL-10. Sixty minutes after a single oral administration of vehicle (10 ml/kg), or paroxetine (30 mg/kg), LPS (0.5 mg/kg, i.p.) or saline (10 ml/kg) was injected. Blood sample was collected 90 min after LPS injection. Serum levels of TNF-α and IL-10 were measured with ELISA. A: Schedule of treatment and sample collection, B: TNF-α, C: IL-10. Data are shown the mean ± S.E.M. (n = 8 for TNF-α and IL-10). ***p < 0.001 compared with LPS-treated group.

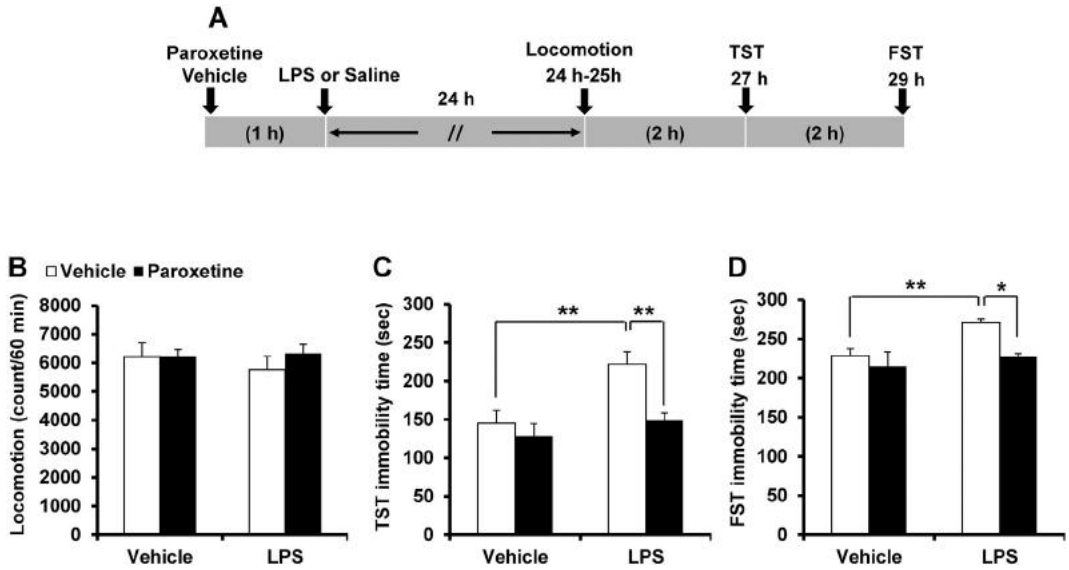

Fig. 4. Antidepressant effects of paroxetine on LPS-induced depression-like behaviors in mice. Paroxetine (30 mg/kg) or vehicle (10 ml/kg) was administered orally 60 min before LPS (0.5 mg/kg, i.p.) administration. Behavioral evaluations were performed 24 h after LPS injection. A: Schedule of treatment and behavioral tests. On day 1, vehicle or paroxetine was orally administered. One hour later, saline or LPS was administered i.p. into mice. On day 2, all behavioral tests were performed in the following order: locomotion on 9:00 AM (24 h after LPS injection), tail suspension test (TST) on 14:00 PM (27 h after LPS injection), forced swimming test (FST) on 16:00 PM (29 h after LPS injection). B: locomotion, C: tail-suspension test (TST), D: forced-swimming test (FST). The values represent the mean ± S.E.M. (n = 11–12). $^{*}p < 0.05$, $^{**}p < 0.01$ compared with the vehicle + LPS group.

and that amycenone shows antidepressant effects in LPS-induced depression-like behavior. The anti-inflammatory effect of amycenone on serum TNF-α and IL-10 levels after LPS injection is consistent with the previous results of SSRIs and SNRIs on serum TNF-α and IL-10 levels (Ohgi et al., 2013; this study). Furthermore, pretreatment with amycenone strongly attenuated the increase in the immobility time in the TST and FST after LPS administration.

Peripheral LPS administration induces sickness behaviors that peaks 2–6 h after administration and gradually wanes (Dantzer et al., 2008). Pro-inflammatory cytokine signaling is activated in the brain in response to peripheral LPS injection, and depression-like behavior peak at 24 h post-LPS injection (Dantzer et al., 2008). In this study, we measured serum TNF-α levels in mice 90 min after LPS injection, and we performed the behavioral evaluations 24 h after LPS injection, as reported previously (Ma et al., 2014; Zhang et al., 2015). Amycenone showed an anti-inflammatory effect on the increased serum levels of TNF-α and IL-10 by LPS injection. Furthermore, similar to paroxetine, amycenone showed antidepressant effects on LPS-induced depressive behaviors in mice. These findings suggest that amycenone shows antidepressant effects via anti-inflammatory activity.

Although amycenone has potent anti-inflammatory effects, the underlying precise molecular mechanisms remain to be determined. A recent study showed that benzyl alcohol derivatives from *H. erinaceum* attenuate the LPS-stimulated inflammatory response through the regulation of NF-κB and AP-1 activity in macrophage cells (Noh et al., 2014). Extracts from *H. erinaceum* were also reported to stimulate nerve growth factor (NGF) mRNA synthesis via activation of the c-jun *N*-terminal kinase pathway in 1321N1 human astrocytoma cells (Bredesen, 2014; Mori et al., 2008), indicating a neurotrophic activity of the extracts from *H. erinaceum*. Furthermore, extracts from *H. erinaceum* show anti-cancer, immune-modulating (Wang et al., 2001),hypolipidemic (Yang et al., 2003), antioxidant (Abdullah et al., 2012; Wong et al., 2009), and neuroprotective effects (Mori et al., 2008; Khan et al., 2013). A recent in vivo study showed that *H. erinaceum* shows a strong anti-fatigue activity by decreasing blood lactic acid, superoxide dismutase, glutathione peroxidase, and malondialdehyde, suggesting an application of *H. erinaceum* in the field of sports nutrition (Liu et al., 2015). Nonetheless, further studies on the molecular mechanisms of amycenone are required to completely understand the effects.

Amycenone is a *H. erinaceum* extract standardized to contain hericenones and amyloban. The present study suggests that amycenone has anti-inflammatory and antidepressant effects in LPS-induced model of depression. Unlike to the current antidepressants (e.g., SSRIs and SNRIs), it is known that amycenone does not have adverse effects (Inanaga, 2012). It is, therefore, likely that amycenone may be sale supplement to prevent inflammation-induced depression.

In conclusion, this study shows that amycenone showed antidepressant effects in LPS-induced inflammation model of depression, consistent with the data of paroxetine. Therefore, amycenone may be a potential supplement to prevent inflammation-related depression.

Conflicts of interest

Dr. Cun Zhuang is an employee of Mushroom Wisdom Ltd. (New Jersey, USA) which produces Amyloban®3399. The other authors declare that they have no conflict of interest.

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